SMALL OFFICE NETWORK DESIGN REPORT

Project Title:

Small Office Network Design with VLANs, Subnetting, DHCP, SSH, and ACL

Prepared By:

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Tools Used:

• Cisco Packet Tracer

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OBJECTIVE

To design and implement a secure, efficient, and professional small office network using VLAN segmentation and subnetting of a single Class C network (192.168.1.0/24), along with inter-VLAN routing, DHCP, SSH, and ACL security features.

NETWORK REQUIREMENTS

- **3 Departments:**
  - **Admin**
  - **IT**
  - **Guest**
- **Efficient IP address utilization using subnetting**
- **Full inter-VLAN communication (with controlled restrictions)**
- **Restricted access for the Guest network**
- **Automatic IP assignment using DHCP**
- **Secure remote management via SSH**

NETWORK DESIGN

Topology Description

- **1 Router – Inter-VLAN routing + DHCP server + ACL enforcement**
- **1 Switch – VLAN segmentation**

- **End Devices – PCs assigned to each VLAN**

VLAN Design

VLAN ID	Name	Department
10	ADMIN	Admin
20	IT	IT
30	GUEST	Guest

IP ADDRESSING SCHEME

The 192.168.1.0/24 network is professionally subnetted into /26 blocks for maximum efficiency and future growth.

VLAN	Subnet	Usable Range	Gateway
10	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.1
20	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.65
30	192.168.1.128/26	192.168.1.129 - 192.168.1.190	192.168.1.129
–	192.168.1.192/26	Reserved for future use	–

CONFIGURATION SUMMARY

1. VLAN Configuration

- **Created VLAN 10 (ADMIN), VLAN 20 (IT), and VLAN 30 (GUEST)**
- **Assigned switch access ports to respective VLANs**

2. Trunk Configuration

- **Configured 802.1Q trunk link between switch and router**
- **Enabled VLAN tagging for inter-VLAN routing**

3. Inter-VLAN Routing (Router-on-a-Stick)

Router subinterfaces configured as:

- VLAN 10 → 192.168.1.1
- VLAN 20 → 192.168.1.65
- VLAN 30 → 192.168.1.129

4. DHCP Configuration

Dedicated DHCP pools for each VLAN:

- ADMIN → 192.168.1.0/26
- IT → 192.168.1.64/26
- GUEST → 192.168.1.128/26

Default gateways assigned automatically.

5. SSH Configuration

- Enabled SSH v2 for secure remote access
- Local username/password authentication
- RSA key generation for encryption

6. ACL Configuration

- Extended ACL created on the router
- Blocked Guest VLAN (30) from accessing Admin VLAN (10)
- All other traffic explicitly permitted

SECURITY IMPLEMENTATION

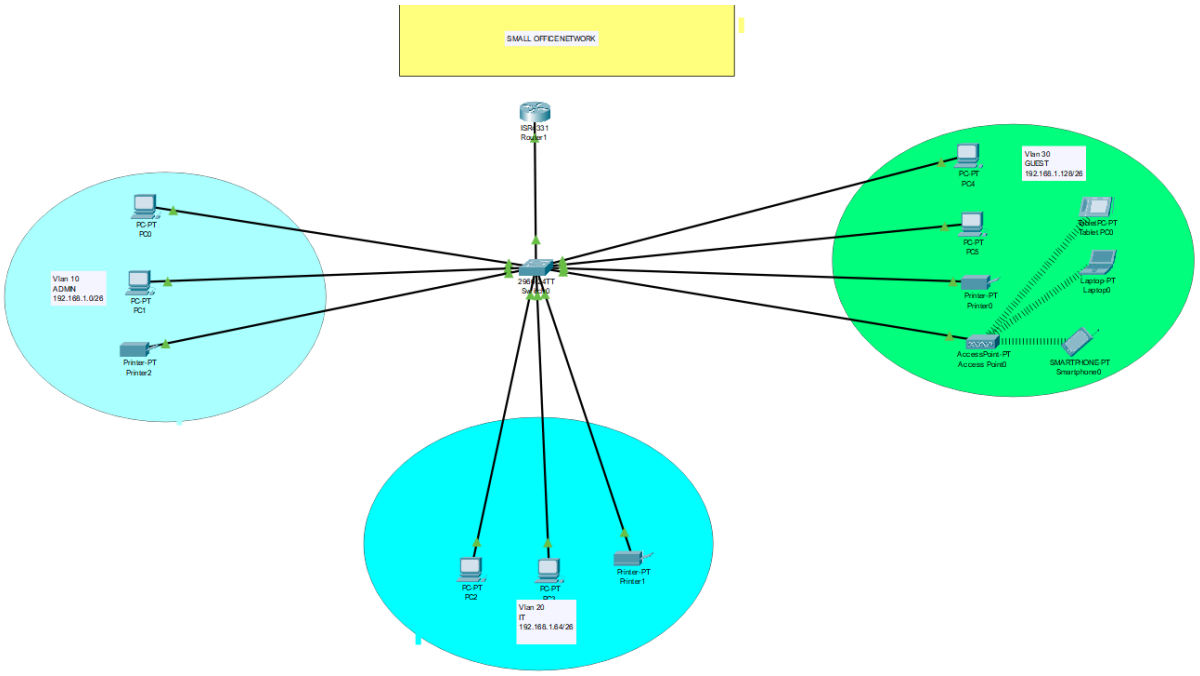
- VLAN segmentation for department isolation
- Precise subnetting for IP efficiency and security
- ACLs for strict access control
- SSH for encrypted management (replacing insecure Telnet)

TESTING & VERIFICATION

Test Case	Result
Admin → IT communication	✅ Success
Guest → Admin access	❌ Blocked
DHCP IP assignment	✅ Working

SSH remote login	✅ **Working**

SCREENSHOTS




```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name GUEST
```

```
Switch(config-vlan)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
```

```
Switch(config-if)#interface fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
```

```
Switch(config-if)#interface fa0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#interface fa0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
Switch(config-if)#interface fa0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
```

```
Switch(config-if)#interface fa0/1
Switch(config-if)#switchport mode trunk
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/0/0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface g0/0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255.255.255.192
Router(config-subif)#
Router(config-subif)#interface g0/0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.1.65 255.255.255.192
Router(config-subif)#
Router(config-subif)#interface g0/0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.1.129 255.255.255.192
```

```
Router(config-subif)#ip dhcp excluded-address 192.168.1.1
Router(config)#ip dhcp excluded-address 192.168.1.65
Router(config)#ip dhcp excluded-address 192.168.1.129
Router(config)#
Router(config)#ip dhcp pool ADMIN
Router(dhcp-config)#network 192.168.1.0 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns 192.168.1.1
Router(dhcp-config)#domain-name Admin.com
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.1.64 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.65
Router(dhcp-config)#dns 192.168.1.65
Router(dhcp-config)#domain-name It.com
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool GUEST
Router(dhcp-config)#network 192.168.1.128 255.255.255.192
Router(dhcp-config)#default-router 192.168.1.129
Router(dhcp-config)#dns 192.168.1.129
Router(dhcp-config)#domain-name Guest.com
```

```
Router(dhcp-config)#hostname OfficeRouter
OfficeRouter(config)#ip domain-name office.local
OfficeRouter(config)#
OfficeRouter(config)#crypto key generate rsa
The name for the keys will be: OfficeRouter.office.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may
take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

OfficeRouter(config)#
*Mar 1 0:2:50.560: %SSH-5-ENABLED: SSH 1.99 has been enabled
OfficeRouter(config)#username admin password 1234
OfficeRouter(config)#
OfficeRouter(config)#line vty 0 4
OfficeRouter(config-line)#login local
OfficeRouter(config-line)#transport input ssh
OfficeRouter(config-line)#
OfficeRouter(config-line)#ip ssh version 2
```

```
OfficeRouter(config)#access-list 100 deny ip 192.168.1.128 0.0.0.63 192.168.1.0 0.0.0.63
OfficeRouter(config)#access-list 100 permit ip any any
OfficeRouter(config)#
OfficeRouter(config)#interface g0/0/0.30
OfficeRouter(config-subif)#ip access-group 100 in
```

```
C:\>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time<1ms TTL=127
Reply from 192.168.1.66: bytes=32 time<1ms TTL=127
Reply from 192.168.1.66: bytes=32 time<1ms TTL=127
```

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.129: Destination host unreachable.
Reply from 192.168.1.129: Destination host unreachable.
Reply from 192.168.1.129: Destination host unreachable.
Reply from 192.168.1.129: Destination host unreachable.
```

CHALLENGES FACED

- **Accurate subnet calculation and avoiding IP overlap**
- **Correct gateway assignment per VLAN**
- **Proper ordering and application of ACL rules**
- **Ensuring seamless trunking and Router-on-a-Stick functionality**

LESSONS LEARNED

- **Practical implementation of /26 subnetting**
- **Real-world application of VLAN segmentation and security**
- **Efficient IP address management**
- **Troubleshooting complex Cisco configurations**

FUTURE IMPROVEMENTS

- **NAT/PAT configuration for internet simulation**
- **Advanced firewall implementation**
- **Layer 3 switching for improved performance**
- **Redundancy (STP, EtherChannel)**

CONCLUSION

This project successfully demonstrates a real-world, production-ready small office network design using industry-standard practices. The combination of VLANs, precise /26 subnetting, DHCP, SSH, and ACLs delivers high security, excellent performance, and efficient resource utilization.

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